Madagascar's endemic freshwater fish face an extinction crisis

Sixty-three percent of Madagascar's ~79 endemic freshwater fish species are threatened with extinction, making them the island's most imperiled vertebrate group. Three species are already extinct, at least 11 Critically Endangered species lack any captive safety net, and invasive species have replaced native fish across most of the Central Highlands and western lowlands. Yet a fragile but growing conservation network — spanning Malagasy researchers, European zoos, hobbyist breeders, and a handful of in-country projects — is fighting to prevent further losses. This report synthesizes the current state of taxonomy, threats, ex-situ programs, in-country efforts, and stakeholder recommendations across all known Malagasy endemic freshwater fish families and genera.

Complete taxonomic inventory reveals extraordinary endemism and alarming threat levels

Madagascar's freshwater fish fauna includes **173 species recorded from freshwater habitats**, of which 123 are exclusively freshwater. Approximately **79 described species are endemic** (found nowhere else on Earth), belonging to two endemic families (Bedotiidae and Anchariidae) and several endemic genera within other families. The true number is higher: at least 15–20 undescribed species have been identified through genetic and morphological work but await formal description.

The IUCN Red List status breakdown for the ~79 endemic species is stark:

IUCN Category	Count	Notable species
Extinct (EX)	2	*Ptychochromis onilahy*, *Pantanodon madagascariensis*
Presumed Extinct (NE)	1	*Malagodon honahona* (described 2024)
Critically Endangered (CR)	~14	*Bedotia tricolor*, *Rheocles derhami*, *R. lateralis*, *Paretroplus dambabe*, *P. gymnopreopercularis*, *P. maculatus*, *P. menarambo*, *Ptychochromis insolitus*, *Ptychochromoides betsileanus*, *Ptychochromoides itasy*, *Typhleotris mararybe*, *T. pauliani*, *Arius festinus*, *A. uncinatus*
Endangered (EN)	~25	Across all families
Vulnerable (VU)	~6	*Paratilapia polleni*, *Paretroplus damii*, *P. kieneri*, *Bedotia masoala*, *Pachypanchax arnoulti*, *Kuhlia sauvagii*
Data Deficient (DD)	~12	Including all 3 *Gogo* catfishes, 4 *Ptychochromis*, 2 *Rheocles*
Not Evaluated	Several	Newly described/undescribed species

Family-by-family breakdown

Bedotiidae (Madagascar rainbowfishes) — the island's only endemic fish family alongside Anchariidae — comprises two genera: *Bedotia* (9 described species, plus 10+ undescribed lineages assessed by IUCN) and *Rheocles* (7 described species). All inhabit eastern forest streams and are acutely sensitive to sedimentation

and canopy loss. *Bedotia tricolor* is the sole Critically Endangered *Bedotia*, restricted to the Manakara River drainage. *Rheocles derhami* (CR), not recorded from 2013 until its **dramatic 2024 eDNA rediscovery** in the Amboaboa River basin, exemplifies how little is known about even described species.

Cichlidae (Malagasy endemic cichlids) represent the greatest diversity, with **~29 species across 6 genera**. The genus *Paretroplus* (13 species, all northwestern/western) includes four Critically Endangered species and the flagship *P. menarambo* — once classified as Extinct in the Wild until rediscovered at Lake Tseny in 2010. *Ptychochromis* (10 species) includes the extinct *P. onilahy* (not seen since 1962) and the iconic *P. insolitus* (Mangarahara cichlid), reduced to 18 wild individuals before rescue in 2013. Monotypic genera *Oxylapia* and *Katria* are confined to the Nosivolo River, while *Ptychochromoides* (3 species) includes two Critically Endangered highland cichlids. *Paratilapia polleni* is likely a species complex of 3–5 cryptic species requiring taxonomic revision.

Aplocheilidae (killifishes) — genus *Pachypanchax* — includes 6 described Malagasy species, all Endangered or Vulnerable. Molecular work indicates the genus needs partial taxonomic revision, as *P. omalonotus*, *P. patriciae*, and *P. varatraza* are not reciprocally monophyletic. At least three additional undescribed species exist.

Anchariidae (Malagasy catfishes) — the second endemic family — contains *Ancharius* (2 species: *A. fuscus* LC, *A. griseus* EN) and *Gogo* (4 species, 3 Data Deficient). The micro-endemic *Gogo* species are among the least-studied Malagasy fish.

Other endemic taxa include the cave-dwelling *Typhleotris* (3 blind cave gobies in southwestern karst systems, 2 CR), the extinct Pantanodontidae (*Malagodon madagascariensis* EX, *M. honahona* described in 2024 and presumed extinct), endemic round herrings (*Sauvagella*, 2 species), freshwater ariid catfishes (*Arius festinus* and *A. uncinatus*, both CR), and several endemic eleotrid and gobiid species.

Recent taxonomic developments (2020–2026)

The most significant recent revision was the **2024 description of *Malagodon honahona*** by Carr, Martin, and Sparks — a new Pantanodontid from southeastern Madagascar's Manombo Reserve, last collected in the late 1990s and presumed extinct. The genus *Malagodon* was erected in 2023 (Meinema & Huber) to separate Malagasy Pantanodontidae from the African *Pantanodon*. The 2024 eDNA survey in the Amboaboa basin identified a likely undescribed *Rheocles* from the Ambodivato River (genetically and morphologically distinct from *R. derhami*) and *Pachypanchax* sp. 'Sofia'. A 2022 DNA barcoding study (Vences et al.) produced **2,860 mitochondrial sequences** forming the first comprehensive barcode library for Malagasy freshwater fish, revealing additional cryptic diversity.

Invasive species and deforestation drive an interconnected crisis

The threats to Malagasy freshwater fish are severe, synergistic, and accelerating. **Invasive species represent the single most pervasive and irreversible threat**, having completely replaced native fish across Madagascar's Central Highlands and much of the west.

Tilapia (*Oreochromis niloticus* and *O. mossambicus*), introduced as food fish from the 1950s onward, now dominate most aquatic systems. In the Amboaboa River basin — the last stronghold of three Critically Endangered species — recent surveys captured **201 *O. mossambicus* and 84 *O. niloticus* versus only 4 *Ptychochromis insolitus* and 1 *Paretroplus gymnopreopercularis***. The **blotched snakehead (*Channa maculata*)**, an air-breathing predator introduced from China in 1976–1978, has devastated native fish populations in most large lakes and is now spreading into previously unaffected streams, including the Namorona River system in Ranomafana National Park. Other established invaders include *Gambusia holbrooki* (which preferentially eats fish larvae over mosquitoes), common carp, goldfish, largemouth bass,

guppies, and swordtails. A recently detected threat is the **marbled crayfish (*Procambarus virginalis*)** — a parthenogenetic species capable of establishing populations from a single individual. The first record of the **Asian fish tapeworm** from Madagascar (2016) was found infecting endemic *Ptychochromis* with high prevalence.

No systematic invasive species removal programs exist for Madagascar's freshwater systems, and no regulations effectively prevent further introductions.

Deforestation from slash-and-burn agriculture (tavy) dumps millions of tonnes of sediment into rivers annually, smothering spawning grounds and raising water temperatures in streams where Bedotia and Rheocles rainbowfishes depend on cool, clear, shaded conditions. Agricultural streams near Ranomafana show **0–14% canopy cover versus 71–83% in adjacent forest streams**, with significantly altered macroinvertebrate communities. Madagascar has lost approximately **90% of its original forest cover**.

Mining poses severe localized threats. The QMM/Rio Tinto ilmenite mine in southeastern Madagascar has caused multiple tailings dam failures (2010, 2018, 2022), resulting in massive fish kills and water contamination with uranium levels **50 times WHO guidelines**. The Ambatovy nickel-cobalt mine in the east has cleared **2,000+ hectares** of rainforest, and pipeline ruptures (most recently November 2024) release pollutants into river systems. Artisanal gold mining introduces an estimated **18 tonnes of mercury per year** into waterways. Sapphire rushes bring thousands of miners into remote areas, increasing sedimentation and habitat disturbance.

Dam construction and water diversion have caused catastrophic local extinctions. The drying of the Mangarahara River due to dam construction and rice irrigation directly caused the functional extinction of *P. insolitus*, *P. gymnopreopercularis*, and *P. nourissati* in that system. **Climate change** compounds all threats through intensifying drought in the south (the worst in 40+ years during 2019–2022), more powerful cyclones, and reduced river flow during dry seasons.

The ex-situ conservation gap: 31 threatened species have no captive safety net

The landmark **Leiss et al. (2022) paper in *Zoo Biology*** — one of the journal's most-cited papers of 2022–2024 — documented that only **19 of 50 threatened endemic species** were held in zoos worldwide (22 including private breeders), with just 9 having reproduced in captivity in the preceding year. **31 threatened species, including 11 Critically Endangered, have zero captive populations.** Total zoo holdings comprised 4,267 individuals globally (3,602 in Europe, 659 in North America), heavily skewed toward the Least Concern *Ptychochromis grandidieri* (1,290 individuals).

The paper's core recommendations remain the field's strategic framework:

- Distribute offspring among more institutions, including qualified private breeders
- Strengthen connections between field research and breeding programs
- Facilitate legal capture and transfer of threatened taxa for ex-situ programs
- Develop integrated in-situ/ex-situ strategies following the IUCN One Plan Approach
- Build ex-situ facilities within Madagascar
- Conduct urgent IUCN Red List reassessments

Cologne Zoo has emerged as the **central European hub** for Malagasy freshwater fish breeding under Prof. Thomas Ziegler. Holding 9 endemic species, the zoo achieved **first European breeding of *Ptychochromis insolitus*** and first-ever European institutional holding of *P. loisellei*, distributing **300+ offspring to 10+ European zoos** including Rotterdam, ZieZoo, and Plock. All nine species were DNA-barcoded to confirm identity for potential repatriation. **Toronto Zoo** played the pivotal role in *P. insolitus* conservation, producing

thousands of hatchlings since its first successful breeding in September 2016 and seeding the European network by transferring stock to Cologne in 2019. **Denver Zoo**, which initiated its Madagascar program in the early 1990s under Alex Saunders, maintains one of North America's most diverse collections (5 rainbowfish, 4 cichlid, 3 killifish species) and co-led the **2010 rediscovery of *Paretroplus menarambo*** at Lake Tseny. ZSL London Zoo runs the **Fish Net Madagascar** program and received *P. insolitus* specimens for breeding. The Aquarium Tropical du Palais de la Porte Dorée (Paris), under Charles Fusari, maintains Malagasy species and leads field research.

Citizen Conservation (Germany), directed by Björn Encke, bridges the gap between zoos and private hobbyists through structured breeding programs. Active CC programs exist for *P. insolitus*, *P. loisellei*, *Bedotia madagascariensis*, and *Paretroplus damii*, with published breeding guidelines and a "Wild at Home" coordination platform. This directly implements the Leiss et al. recommendation to include private breeders as conservation partners.

The **AZA Freshwater Fishes TAG** oversees Regional Collection Plans for North American institutions. ZIMS data show **16 Malagasy freshwater fish species (659 individuals)** across AZA facilities, with 6 species successfully reproducing. The **EAZA Freshwater Teleost TAG**, chaired by Brian Zimmerman (now at Bristol Zoological Society), coordinates European EEP programs, with Peter Petersen at Den Blå Planet managing the EAZA Cichlid EEP.

Species with the most critical ex-situ gaps

Among the 11 Critically Endangered species absent from captivity, the highest priorities include *Rheocles derhami* (rediscovered 2024 but no captive program initiated), *Bedotia tricolor* (micro-endemic, SE Madagascar), *Ptychochromoides betsileanus* and *P. itasy* (highland cichlids facing tilapia invasion), *Typhleotris mararybe* and *T. pauliani* (cave fishes with extraordinary phylogenetic importance), and *Arius festinus* and *A. uncinatus* (Critically Endangered freshwater catfishes from the northwest). No formal **CPSG Ex-situ Conservation Assessment (ECA)** workshop has been conducted specifically for Malagasy freshwater fishes — a significant gap identified by multiple researchers.

CARES, SHOAL, and the IUCN FFSG anchor the global coordination architecture

CARES Preservation Program

Madagascar holds the **second-largest representation on the CARES priority list** (66 species, 15% of the global total), behind only Mexico. CARES, founded in 2004 and coordinated by Claudia Dickinson with the late Paul Loiselle serving as Technical Editor and Madagascar Regional Coordinator, encourages hobbyists to maintain threatened species. The CARES Bedotiidae list alone includes 22 species, while Malagasy cichlids are well-represented (*Katria katria*, *Oxylapia polli*, all *Paretroplus*, *Ptychochromis insolitus*, *P. loisellei*, and multiple *Pachypanchax*). A 2019 academic analysis found that approximately two-thirds of CARES species also appear on the IUCN Red List, though some classification discrepancies exist.

SHOAL Conservation and the 1,000 Fishes Blueprint

SHOAL, launched in 2019 under Executive Director Mike Baltzer, released its flagship **1,000 Fishes Blueprint in April 2024** — a framework for halting extinctions and recovering populations of 1,000 of Earth's most threatened freshwater fish by 2035, drawn from a priority list of 2,300+ species developed with the IUCN FFSG and IUCN Biodiversity Assessment and Knowledge Team. **Madagascar's *Typhleotris pauliani*** is explicitly highlighted as a priority species, and *Ptychochromoides itasy* appears among SHOAL's "Top 10 Most Wanted Lost Fishes" campaign (co-led with Re:wild). SHOAL operates through three pillars — Inspire, Build, and Act — and established the **SHOAL Alliance of Zoos and Aquaria** in late 2023. Specific

Madagascar Local Action Partners have not been publicly identified, but SHOAL's network includes key figures with active Madagascar field experience, including Brian Zimmerman (FFSG Co-Chair) and Jörg Freyhof.

IUCN SSC Freshwater Fish Specialist Group (FFSG)

The FFSG Madagascar Region was co-chaired by **Paul Loiselle** and **Dr. Tsilavina Ravelomanana** (University of Antananarivo). The landmark 2004 IUCN Red List Assessment found 22% of Madagascar's freshwater fish Critically Endangered and 34% Endangered. Many assessments are now outdated; Leiss et al. (2022) and Ziegler et al. (2020) recommended specific upgrades (*Pachypanchax sakaramyi* from EN to CR; *Gogo atratus* from DD to VU) and potential downlistings (*Bedotia leucopteron*, *B. madagascariensis*). **No comprehensive global reassessment** of all Malagasy freshwater fish has been completed in the 2022–2026 period, though individual species receive rolling updates.

Regarding **IUCN Resolution 112** (proposed by the New Mexico BioPark Society concerning ex-situ management of freshwater fishes): the specific resolution text could not be confirmed in publicly available sources. The ABQ BioPark's Center for Species Survival, established 2018 in partnership with the IUCN SSC, has contributed to 2,000+ species-level assessments spanning 96 countries, with Tim Lyons serving as Species Survival Officer for aquatic habitats. Their primary freshwater fish work has focused on Mexico and Central America rather than Madagascar specifically.

The informal "Madagascar Freshwater Fish Conservation Group"

No formal standalone organization exists under this name. Instead, the conservation landscape operates as an **interconnected network** through the IUCN FFSG Madagascar Region, the Madagascar Fauna and Flora Group (MFG), the CARES program, and an informal ex-situ breeding network coordinated by Ziegler (Cologne), Fusari (Paris), and colleagues. Key members include Andolalao Rakotoarison and Tsilavina Ravelomanana (University of Antananarivo), Miguel Vences (Braunschweig), John Sparks and Melanie Stiassny (AMNH), and Paul Reinthal (University of Arizona). Their collective published output — particularly Leiss et al. (2022), Vences et al. (2022), and Fusari/Carvalho et al. (2024) — effectively constitutes the group's strategic plan and action framework.

In-country conservation efforts center on three critical river systems

The Amboaboa River basin: Fish Net Madagascar

Fish Net Madagascar, coordinated by Charles Fusari, is the most comprehensive in-country freshwater fish program, focused on the **Amboaboa River basin** in the Sofia drainage of northeastern Madagascar — the last known watershed for three Critically Endangered species (*P. insolitus*, *P. gymnopreopercularis*, *R. derhami*). The program operates on multiple fronts: regular field monitoring along a 75-km stretch where the 5 focal species occupy only ~24 km during dry season; **translocation planning** for *P. insolitus* upstream into the Réserve Spéciale de Marotandrano (with a ZSL-protocol disease risk analysis underway and temperature tolerance studies completed in 2023); and **community pond management** through the Association des Producteurs Privés d'Alevins d'Andapa (APPA), led by Guy Tam Hyock, which maintains backup populations of endemic species in ponds near Andapa. Madagascar National Parks rangers from Marotandrano conduct monthly water quality monitoring.

The **September 2024 *Scientific Reports* paper** (Oliveira Carvalho, Fusari, et al.) represents the program's most significant recent output. Using combined eDNA metabarcoding and traditional fishing across 14 sites, the team **rediscovered *Rheocles derhami*** (331 individuals captured, none seen since 2013), confirmed *P. insolitus* (N=4), and detected 19 total species including multiple invasive aliens. The study generated **12S**

reference barcodes for 10 species — the region's most comprehensive molecular dataset — and identified likely undescribed species (*Rheocles* sp. 'Ambodivato' and *Pachypanchax* sp. 'Sofia').

Partners include Aquarium Tropical du Palais de la Porte Dorée (Paris), ZSL, Toronto Zoo, University of Antananarivo, University of Oslo (eDNA expertise, Quentin Mauvisseau), Madagasikara Voakajy (community engagement under Jelot Hernandez), Cologne Zoo, and Citizen Conservation. Funding comes from National Geographic, EUAC, the American Cichlid Association, and partner zoos.

The Nosivolo River: Durrell's community-based model

The **Nosivolo River** in east-central Madagascar hosts the highest concentration of endemic freshwater fish on the island — **19 endemic species, 4 found nowhere else** — including the only known habitat of *Oxylapia polli* (Songatana cichlid) and *Katria katria*. Durrell Wildlife Conservation Trust, working since 2003 with Conservation International, helped establish Madagascar's **first river Ramsar Site (2010)** here. Community-based fisheries management has been remarkably successful: riverine communities maintain strict fishing closures, enforce closed seasons of **4–5 months** (exceeding legal requirements), and impose mesh-size regulations stricter than national law. Joint control teams of community representatives and forestry officers conduct patrols. Alternative livelihoods (farming, beekeeping, artisanal work) supported by micro-financing have reduced fishing pressure. Reports indicate endemic fish are **more numerous and larger** than before conservation measures began.

Lake Tseny and northwestern Madagascar

Lake Tseny in the Sofia River basin harbors the only known wild population of *Paretroplus menarambo* (Pinstripe Damba), rediscovered in 2010 by a Denver Zoo/Madagasikara Voakajy team. The Malagasy NGO **Tany Maeva** works with Toronto Zoo and Denver Zoo on community-based protection. The **Aspinall Foundation**, funded through Fondation Segré and IUCN Save Our Species, conducts the most active on-the-ground conservation for *Paretroplus maculatus* in the Maevatanana/Ambato-Boeni region, implementing community-based management and fishing regulations. A concerning October 2024 field expedition reported the **apparent extirpation of *P. maculatus* from Lake Ravelobe**, suggesting continued decline despite efforts.

The Re:wild Makira expedition (September 2023)

Re:wild's first multi-taxa lost species expedition to **Makira Natural Park** (Madagascar's largest intact forest, managed by WCS) achieved a remarkable result: all **three target freshwater fish species were rediscovered** — *Bedotia alveyi* (lost since 2003), an undescribed *Rheocles* sp. (lost since 2006), and *Ptychochromis makira* (lost since 2003). University of Antananarivo fish biologists Tsilavina Ravelomanana and Fetra Andriambelomanana led the fish team, and Christina Biggs' eDNA sampling detected 37 additional vertebrate species not observed during visual surveys.

Other in-country work

Durrell's **Lake Alaotra** program, primarily focused on the Alaotran gentle lemur, has planted **160,000+ seedlings to restore 127 hectares**, reducing erosion benefiting *Rheocles alaotrensis* and other endemic fish. The **Madagascar Biodiversity Partnership** operates aquaponic units with *Paratilapia* specimens but freshwater fish conservation is not its primary focus. **GERP** and the **Tropical Biodiversity Alliance** have no documented freshwater fish programs. **No formal national conservation action plan** specifically for endemic freshwater fish exists — the most comprehensive strategic recommendations come from the published literature and the SHOAL Blueprint.

Paul Loiselle's death leaves an irreplaceable gap

Dr. Paul V. Loiselle died on April 15, 2025, aged 79, leaving a profound void in Malagasy freshwater fish conservation. As Curator of Freshwater Fishes at the New York Aquarium (WCS) and later Senior Conservation Fellow, he conducted **14 years of fieldwork in Madagascar**, discovering 15 freshwater fish species. He co-chaired the IUCN FFSG Madagascar Region, served as CARES Technical Editor and Madagascar Regional Coordinator, and was a founding Fellow of the American Cichlid Association. Two species bear his name: *Ptychochromis loisellei* and *Paretroplus loisellei*. He described or co-described *Bedotia leucopteron*, redescribed *B. madagascariensis* and *B. geayi*, co-described *Rheocles vatosoa*, and authored four new *Pachypanchax* species descriptions.

Mongabay's obituary (July 2025) called him **"a bridge builder, between hobbyists and scientists, zoos and field researchers, aquariums and conservationists"** — a characterization that captures his unique role connecting the CARES hobbyist network with professional ichthyology and zoo-based conservation. His keynote address at the 2016 International Aquarium Congress warned of "tinkering with ecology on a Faustian scale." His last known public talk was given in August 2024.

The **ACA Paul V. Loiselle Conservation Fund** (established in his honor before his death) continues as an ongoing memorial, awarding annual grants ($600–$5,413) for cichlid conservation research. Past grants funded collection of founders for *Paretroplus maculatus* and *Ptychochromoides itasy* and fish inventories in southeastern Madagascar. Citizen Conservation in Germany noted the "closing of a circle" — Loiselle began as a private aquarist, and now private aquarists breed species bearing his name. **His successor as FFSG Madagascar Region co-chair has not been publicly announced.**

Funding remains critically insufficient relative to need

Freshwater fish receive disproportionately less conservation funding compared to terrestrial and marine species in Madagascar, despite populations declining twice as fast. The primary funding landscape includes:

- **Critical Ecosystem Partnership Fund (CEPF)**: $12.3 million invested in the Madagascar hotspot between 2015–2022; the EU signed a **€10 million convention** with AFD at COP28 (2023) to continue support. A $150,000 ecosystem profile update is underway for 2026.
- **Mohamed bin Zayed Species Conservation Fund**: Grants up to $25,000 per project; three annual application rounds. Relevant but not Madagascar-fish-specific.
- **ACA Paul V. Loiselle Conservation Fund**: Small but targeted grants specifically for cichlid conservation.
- **SHOAL Conservation**: Fundraises through the "SOS: Support Our SHOAL" campaign bridging hobbyists and conservation.
- **Zoo-based funding**: Toronto Zoo, Cologne Zoo, ZSL, Denver Zoo, and Bristol Zoological Society all provide institutional support for breeding and field programs.
- **World Bank SWIOFish 2**: $74.15 million for southwest Indian Ocean fisheries governance (2017), primarily marine/coastal.
- **GIZ Sustainable Freshwater Aquaculture Development**: Supports rice-fish farming but focused on food security rather than endemic species.

Recent publications (2023–2026) reflect a small but active research community

The most impactful recent publications include:

- **Oliveira Carvalho et al. (2024)**, *Scientific Reports*: eDNA rediscovery of *R. derhami* and multi-method Amboaboa survey
- **Leiss et al. (2022)**, *Zoo Biology*: The ex-situ conservation gap analysis (one of the journal's most cited papers 2022–2024)
- **Vences et al. (2022)**, *PLoS ONE*: DNA barcode library for Madagascar's threatened ichthyofauna

- **Carr, Martin & Sparks (2024)**, *American Museum Novitates*: Description of *Malagodon honahona*
- **Meinema & Huber (2023)**: Erection of genus *Malagodon*
- **SHOAL 1,000 Fishes Blueprint (2024)**: Global conservation framework including Malagasy priorities
- **Re:wild Makira expedition report (July 2024)**: Rediscovery of three lost freshwater fish species

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Conclusion: What must happen next

The conservation outlook for Malagasy freshwater fish is precarious but not hopeless. The 2024 rediscoveries of *Rheocles derhami*, *Bedotia alveyi*, and *Ptychochromis makira* demonstrate that species presumed lost can persist in remote refugia. The expanding ex-situ network — from Toronto and Cologne zoos to Citizen Conservation's private breeders — provides insurance against catastrophic in-situ losses. Community-based management on the Nosivolo River proves that local stewardship can reverse declines.

Five actions would transform the field's trajectory. **First**, a formal **CPSG Ex-situ Conservation Assessment workshop** specifically for Malagasy freshwater fish is urgently needed to systematically prioritize the 31 threatened species lacking captive populations. **Second**, the **translocation of *P. insolitus* into the Marotandrano Reserve** — years in planning — must be executed while the disease risk analysis is completed. **Third**, the **IUCN must conduct comprehensive Red List reassessments** for all Malagasy freshwater fish; many assessments are over a decade old, and several species (including 12 Data Deficient taxa) lack any meaningful evaluation. **Fourth**, the death of Paul Loiselle demands a deliberate **succession strategy** for FFSG Madagascar leadership and the institutional knowledge he carried. **Fifth**, freshwater fish must be explicitly integrated into Madagascar's **national protected area strategy and aquaculture development plans**, which currently focus on marine resources and food production while ignoring the island's most threatened vertebrate group.

Madagascar's endemic freshwater fish represent **irreplaceable evolutionary heritage** — basal lineages whose closest relatives live thousands of miles away, cave fish of extraordinary phylogenetic significance, and a radiation of cichlids found nowhere else on Earth. The scientific community, zoo network, and hobbyist breeders have built a foundation. Whether it proves sufficient depends on whether funding, policy, and institutional will can match the scale of the crisis before more species follow *Ptychochromis onilahy* into extinction.